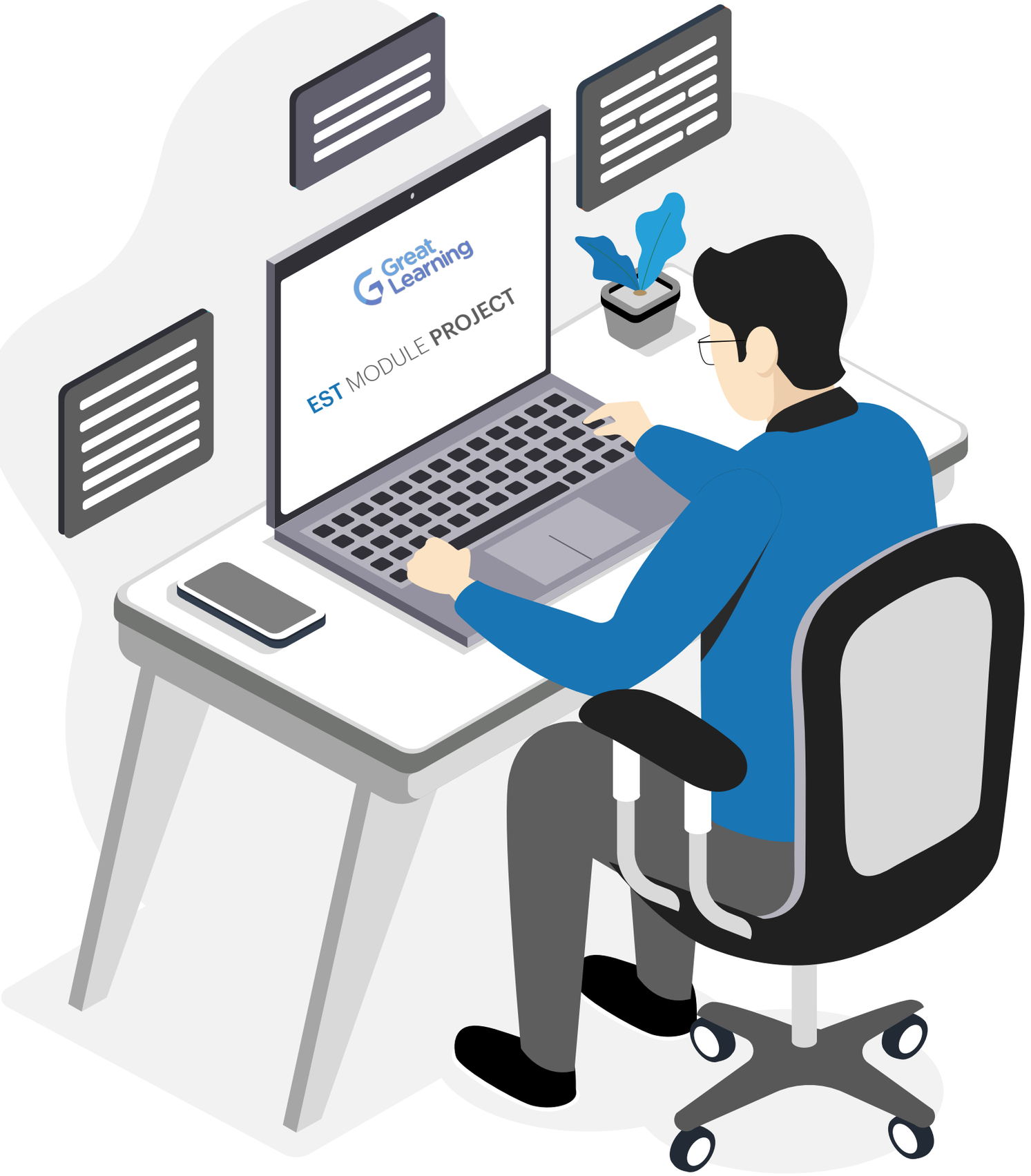


AIML | MODULE PROJECT

©Great Learning. Proprietary content. All Rights Reserved. Unauthorised use or distribution prohibited

Ensemble Techniques

TOTAL SCORE 60

General Instructions:

1. Submission of all the parts is expected in 1 notebook only
2. Expected submission format: 1 '.ipynb' notebook and 1 '.html' notebook only
3. 50% marks will be deducted if insights/steps are missing in the corresponding questions.
4. If output for any code cell is missing, 50% marks will be deducted.
5. Any kind of Plagiarism will lead to 0 (zero) Marks.

Submission Format:

1. '.ipynb' (Jupyter Notebook) and
2. '.html' (Jupyter Notebook > File > Download as > HTML)
5 Marks will be deducted if submission in any of the formats is missing.

Project - 60 Marks

- **DOMAIN:** Telecom
- **CONTEXT:** A telecom company wants to use their historical customer data to predict behaviour to retain customers. You can analyse all relevant customer data and develop focused customer retention programs.
- **DATA DESCRIPTION:** Each row represents a customer, each column contains customer's attributes described on the column Metadata. The data set includes information about:
  - Customers who left within the last month – the column is called Churn
  - Services that each customer has signed up for – phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies
  - Customer account information – how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges
  - Demographic info about customers – gender, age range, and if they have partners and dependents

- **PROJECT OBJECTIVE:** To Build a model that will help to identify the potential customers who have a higher probability to churn. This helps the company to understand the pinpoints and patterns of customer churn and will increase the focus on strategizing customer retention.

- **STEPS AND TASK [60 Marks]:**

1. **Data Understanding & Exploration: [10 Marks]**
   - A. Read 'TelcomCustomer-Churn_1.csv' as a DataFrame and assign it to a variable. [2 Mark]
   - B. Read 'TelcomCustomer-Churn_2.csv' as a DataFrame and assign it to a variable. [2 Mark]
   - C. Merge both the DataFrames on key 'customerID' to form a single DataFrame [4 Mark]
   - D. Verify if all the columns are incorporated in the merged DataFrame by using simple comparison Operator in Python. [2 Marks]

2. **Data Cleaning & Analysis: [30 Marks]**
   - A. Impute missing/unexpected values in the DataFrame. [4 Marks]
   - B. Make sure all the variables with continuous values are of 'Float' type. [4 Marks]
     [For Example: MonthlyCharges, TotalCharges]
   - C. Using the DataFrame obtained from previous step, plot two suitable charts and pivot table for all the Categorical features. [8 Marks]
   - D. Share insights for Q2.c. [4 Marks]
   - E. Encode all the appropriate Categorical features with the best suitable approach. [4 Marks]
   - F. Split the data into 80% train and 20% test. [2 Marks]
   - G. Normalize/Standardize the data with the best suitable approach. [4 Marks]

3. **Model building and Improvement: [20 Marks]**
   - A. Train a model using Gradient Boost and use RandomizedSearchCV to train on best parameters. Also print best performing parameters along with train and test performance. [10 Marks]
   - B. Train a model using Gradient Boost and use GridSearchCV to train on best parameters. Also print best performing parameters along with train and test performance. [10 Marks]

©Great Learning. Proprietary content. All Rights Reserved. Unauthorised use or distribution prohibited